



Residential exposure to petrochemical industrial complexes and the risk of leukemia: A systematic review and exposure-response meta-analysis[☆]

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ABSTRACT

Exposure to chemicals produced by petrochemical industrial complexes (PICs), such as benzene, ionizing radiation, and particulate matters, may contribute to the development of leukemia. However, epidemiological studies showed controversial results. This systematic review and meta-analysis aimed to summarize the association between residential exposure to PICs and the risk of leukemia incidence, focusing on exposure-response effects. We searched PubMed, Embase, Web of Science, and Cochrane Library databases for studies published before September 1st, 2019. Observational studies investigating residential exposure to PICs and the risk of leukemia were included. The outcome of interest was the incidence of leukemia comparing to reference groups. Relative risk (RR) was used as the summary effect measure, synthesized by characteristics of populations, distance to PICs, and calendar time in meta-regression. We identified 7 observational studies, including 2322 leukemia cases and substantial reference groups, in this meta-analysis. Residential exposure to PICs within a maximal 8-km distance had a 36% increased risk of leukemia (pooled RR = 1.36, 95% CI = 1.14–1.62) compared to controls, regardless of sex and age. In terms of leukemia subtypes, residential exposure to PICs was associated with the risks of acute myeloid leukemia (AML, pooled RR = 1.61, 95% CI = 1.12–2.31) and chronic lymphocytic leukemia (CLL, pooled RR = 1.85, 95% CI = 1.11–6.42). In meta-regression, the positive association occurred after 10 years of follow-up with a pooled RRs of 1.21 (95% CI = 1.02–1.44) and then slightly increased to 1.77 (95% CI = 1.35–2.33) at 30 years after follow-up. No effect modification was found by sex, age, and geographic locations.

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1. Introduction

Petrochemical industrial complexes (PICs) are clusters of industrial plants that manufacture petroleum and chemical products during the process of oil refineries (Division, 2008). Chemicals and toxicants produced by PICs (e.g., benzene, dioxins, heavy metals, and other volatile organic compounds (VOCs)) may contribute to increased risks of various types of cancer (Belson et al., 2006; Nadal

et al., 2011; Kaisarevic et al., 2007; Gariazzo et al., 2005), possibly through the pollution of the air (Cetin et al., 2003), soil (Nadal et al., 2004), and drinking water (Wu et al., 2011). Available epidemiological studies suggested an increased risk of lung cancer among residents living near PICs (Lin et al., 2018a), as well as a higher mortality rate and a shorter life expectancy (Lin et al., 2017a). Of those toxicants, benzene, one notorious chemical of VOCs, is classified as Group 1 carcinogen for hematopoietic cancers, such as leukemia, by the International Agency for Research on Cancer. However, studies investigating the impact of PICs on the incidence of leukemia revealed controversial results (Wilkinson et al., 1999), perhaps due to inadequate study power and/or calendar time-dependent factors.

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Environmental and genetic factors contribute extensively to the development of leukemia (Jin et al., 2016). For example, long-term exposure to either benzene, ionizing radiation, or particulate matters (PM), which are by-products commonly found in PICs, was reported to be associated with leukemia based on different levels of evidence (Filippini et al., 2019). Other studies also found that residents living close to PICs had a higher incidence of leukemia (Straif AC and Samet, 2013; Schuz and Erdmann, 2016; Loomis et al., 2013). Biologically, benzene could target critical genes in carcinogenic pathways and induce genetic or epigenetic alterations in hematopoietic stem cells (McHale et al., 2012), suggesting a possible explanation for the development of leukemia.

However, current evidences on the association between residential exposure to PICs and the incidence of leukemia remained inconclusive. Yu et al. found an increased incidence of leukemia among young residents exposed to PICs in Taiwan (Yu et al., 2006a), while Pekkanen et al. did not observe the association in a population study in Finland (Pekkanen et al., 1995a). The potential discrepancy of results between studies may come from different characteristics across populations, definition of exposed cases, exposure duration, and calendar time affecting the law regulation to PICs. Hence, we aimed to conduct a systematic review and meta-analysis to summarize the association between exposure to PICs and the risk of leukemia, focusing on different characteristics of populations and exposure-response effects across various time periods.

2. Methods

2.1. Protocol and registration

The methodology of systematic review and meta-analysis followed the Preferred Reporting Items for Systemic review and Meta-Analysis Protocols (PRISMA-P) (Moher et al., 2015) and Cochrane handbook (Higgins and Green, 2011). This study was registered in the PROSPERO international prospective register of systematic reviews (Registration number: CRD42019132581).

2.2. Literature search and selection criteria

We searched databases, including PubMed, Embase, Web of Science, ScienceDirect, and Cochrane Library, for original studies published before September 1st, 2019, by three independent researchers (BP, YH, and KB). The keywords and Medical Subject Heading (MeSH) terms included ("Refinery OR Petroleum OR Petrochemical OR Oil and Gas Industry) AND (Leukemia OR Leukemia OR AML OR CML OR ALL OR CLL OR multiple myeloma)" during searching process. We did not restrict the language and types of studies.

Studies were included based on *a priori* criteria as follows (Division, 2008): using residential distance to PICs as the proxy of the exposure of interest (Belson et al., 2006); using leukemia incidence as the outcome of interest; and (Nadal et al., 2011) consisting of retrievable effect measure estimates. Studies not using residential distance to PICs as the proxy of exposure were excluded. The outcome of interest was determined by either International Classification of Diseases (ICD) codes or pathological confirmation. For studies reporting several estimates by lengths of follow-up, we chose the estimate with the longest follow-up period.

2.3. Data extraction

The data were extracted by two independent reviewers (BP and KB). Among 7 included studies, subjects enrolled from different age categories and sexes were considered as distinctive study

populations (a total of 13 study populations). Basic characteristics of each study population were extracted, including sample size, age, sex, geographical locations, PICs operation period, follow-up period, methodology for outcome and exposure measurements, and chemicals manufactured in the PICs. Four types of effect estimates in those studies were extracted: odds ratio (OR), relative risk (RR), standardized incidence ratio (SIR) and age-standardized rate ratio (ASRR).

2.4. Statistical analysis

We chose random effect models to estimate the summary measures for this meta-analysis. Of the effect measures, SIR and ASRR were age- and sex-standardized rate ratios representing the comparison between observed and expected incidence rates of the exposed group (Lin et al., 2017b). We approximated and applied OR, SIR, and ASRR as relative risk (RR) to summarize the pooled effect estimates, assuming the occurrence of leukemia is a rare event (Symons and Taulbee, 1981; Viera, 2008). To approximate standard error of natural log of RR, standard errors of logarithm-transformed RR, $SE(\ln RR)$, were calculated from the confidence intervals (Lin et al., 2017b; Lin et al., 2018b). Considering various etiologies of leukemia subtypes, we further examined the relative risks by subtypes of leukemia. To investigate the effect modification by other factors, we performed subgroup analyses with stratification by sex (i.e., males, females, or both), age categories, different effect measures (i.e., RR, OR, SIR/ASRR), and geographic locations (i.e., Asia, North America, and Europe). A Bayesian model was also applied to address potential heterogeneity and the limited number of studies (Gunhan et al., 2019). We further applied meta-regression to investigate the possible origins of heterogeneity (Higgins and Thompson, 2002), including the start- and end-year of PICs operation, the start- and end-year of follow-up, follow-up duration, latency (i.e., the duration between the start-year of PICs operation to the start-year of follow-up), and residential distances to PICs (in kilometers). Heterogeneity was assessed by I^2 statistics, tau statistics and Q test. *P*-values was based on omnibus test following χ^2 distribution (Viechtbauer, 2010). All analyses were conducted using R (version 3.5.2).

2.5. Quality and risk of bias assessment

We chose the Newcastle-Ottawa Quality Assessment Scale (NOQAS) (Wells et al., 2009) to assess the quality of all included case-control and cohort studies, with a maximum of 9 stars containing three metrics (Division, 2008): selection (Belson et al., 2006); comparability; and (Nadal et al., 2011) exposure (case-control) or outcome (cohort). The semi-quantitative NOQAS has great inter-rater reliability (Hootman et al., 2011), explaining its widespread use as a tool in meta-analyses (Hartling et al., 2012).

Sensitivity analysis was performed by leave-one-out method to validate the pooled effect estimates. The normal quantile-quantile plot (Q-Q plot) was used to examine data quality, as well as heterogeneity and publication bias (Bax et al., 2009). Each dot represents a distinctive study population, and dots are expected to be displayed in a straight line if the data is normally distributed (Bax et al., 2008). We performed two other metrics to investigate publication bias: the visual examination of funnel plot analysis and Begg's test for quantitative analysis.

3. Results

After reviewing the full-text articles of 22 studies, we included seven original studies (five cohort studies and two case-control studies) and 13 study populations in this meta-analysis

(Wilkinson et al., 1999; Barregard et al., 2009; Gazdek et al., 2007; Linos et al., 1991; Pekkanen et al., 1995b; Sans et al., 1995; Yu et al., 2006b), consisting of 2322 cases and various reference controls from six countries across North America, Asia, and Europe (Fig. 1). The follow-up period ranged between 1971 and 2000, with a median follow-up duration of 10 years (IQR 6–29 years). The threshold of distance-to-PICs used to define the exposure of interest varied across studies, ranging between 0.8 (Linos et al., 1991) and 8 km (Pekkanen et al., 1995b). Petrochemical substances manufactured in those PICs were similar, including benzene, polycyclic aromatic hydrocarbons (PAH), 1,3-butadiene, and ionizing radiation.

3.1. Main analysis and meta-regression

People living close to PICs (i.e., within a maximum distance of 8 km) had a 36% increased risk of incident leukemia (pooled RR = 1.36, 95% CI = 1.14–1.62), comparing to those living far away (Fig. 2A). The result is consistent in a Bayesian model analysis (RR = 1.37, 95% prior distribution = 1.11–1.70). We found that the summary RRs of leukemia varied across different follow-up durations, which correlates to the length of exposure. Considering different follow-up durations in the meta-regression, a positive association occurred after 10 years of follow-up with a pooled RR of 1.21 (95% CI = 1.02–1.44) and slightly increased onward (Fig. 2B), suggesting a period of follow-up is probably required to have a significant effect (Table 1).

The calendar time frames of the PICs operation (starting periods between 1955 and 1981 and ending periods between 2000 and 2019) and the follow-up durations of included studies were illustrated in Fig. 3A. Regarding the start-year of PICs operation between 1964 and 1990, meta-regression showed a significantly positive

association between exposure to PICs and the risk of leukemia (pooled RR from 1.22 to 2.15), with an average of 2% increase of pooled RR (slope = 1.02, 95% CI = 1.00–1.04) annually, comparing to the prior start-year (Fig. 3B). In terms of the end-year of PICs operation from 2000 to 2018, on the contrary, although the positive association persisted across calendar year (pooled RR from 1.80 to 1.03), there was a 3% decrease of pooled RR (slope = 0.97, 95% CI = 0.96–0.98) per year (Fig. 3C). Other factors, such as start- and end-year of follow-up, latency, and residential distances from PICs, did not significantly affect the heterogeneity of the summary estimates.

3.2. Subgroup analysis

Subgroup analysis did not find significant effect modifiers affecting the summary relative risks of leukemia. The summary effects differed by subtypes of leukemia, particularly higher in acute myeloid leukemia (AML, pooled RR = 1.61, 95% CI = 1.12–2.31), chronic lymphoblastic leukemia (CLL, pooled RR = 1.85, 95% CI = 1.01–6.42) and acute lymphoblastic leukemia (ALL, pooled RR = 1.50, 95% CI = 0.97–1.69). In addition, the summary risk of leukemia was slightly higher for males (pooled RR = 2.46, 95% CI = 1.22–4.97) versus females (pooled RR = 1.88, 95% CI = 1.32–2.67) (Table 2). Regarding age, residents aged below 30 showed a lower, but not significant, RR (RR = 1.22, 95% CI = 0.96–1.56) than the whole population (RR = 1.43, 95% CI = 1.15–1.79). Considering the geographic locations, the summary RRs were similar, ranging from 1.13 (95% CI = 0.92–1.40), 1.25 (95% CI = 0.96–1.64), and 1.50 (95% CI = 1.15–1.96) in North America, Asia, and Europe, respectively (Table 2).

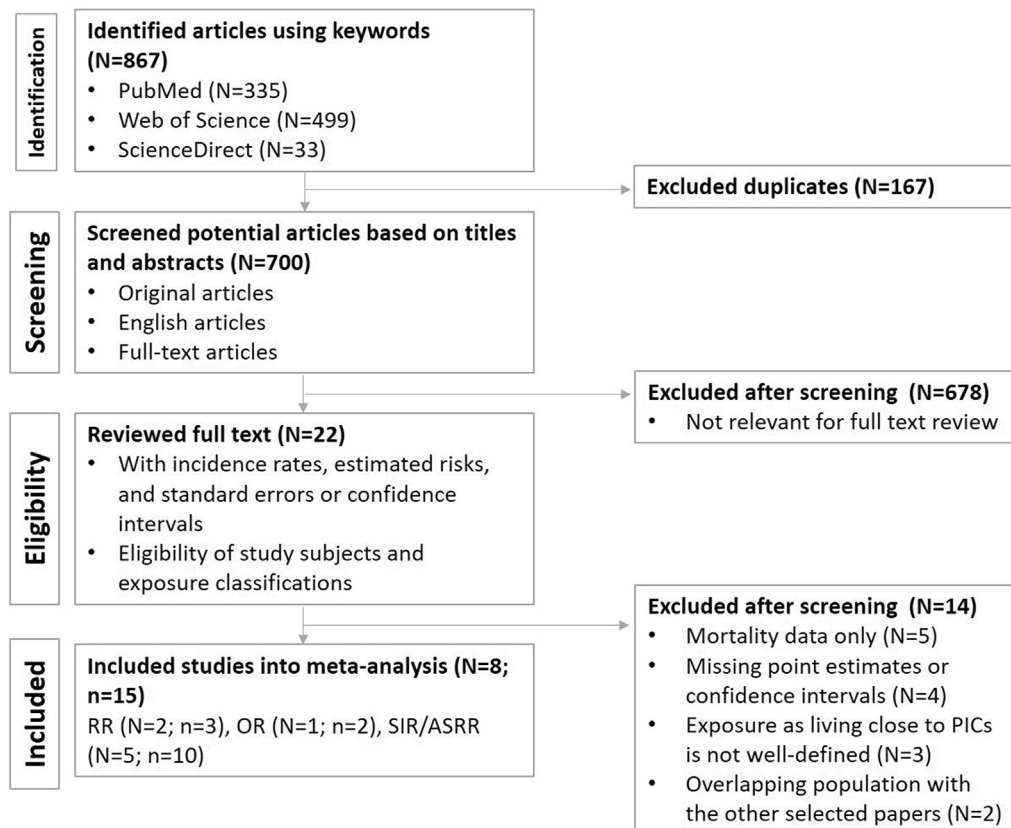


Fig. 1. The flowchart of study enrollment.

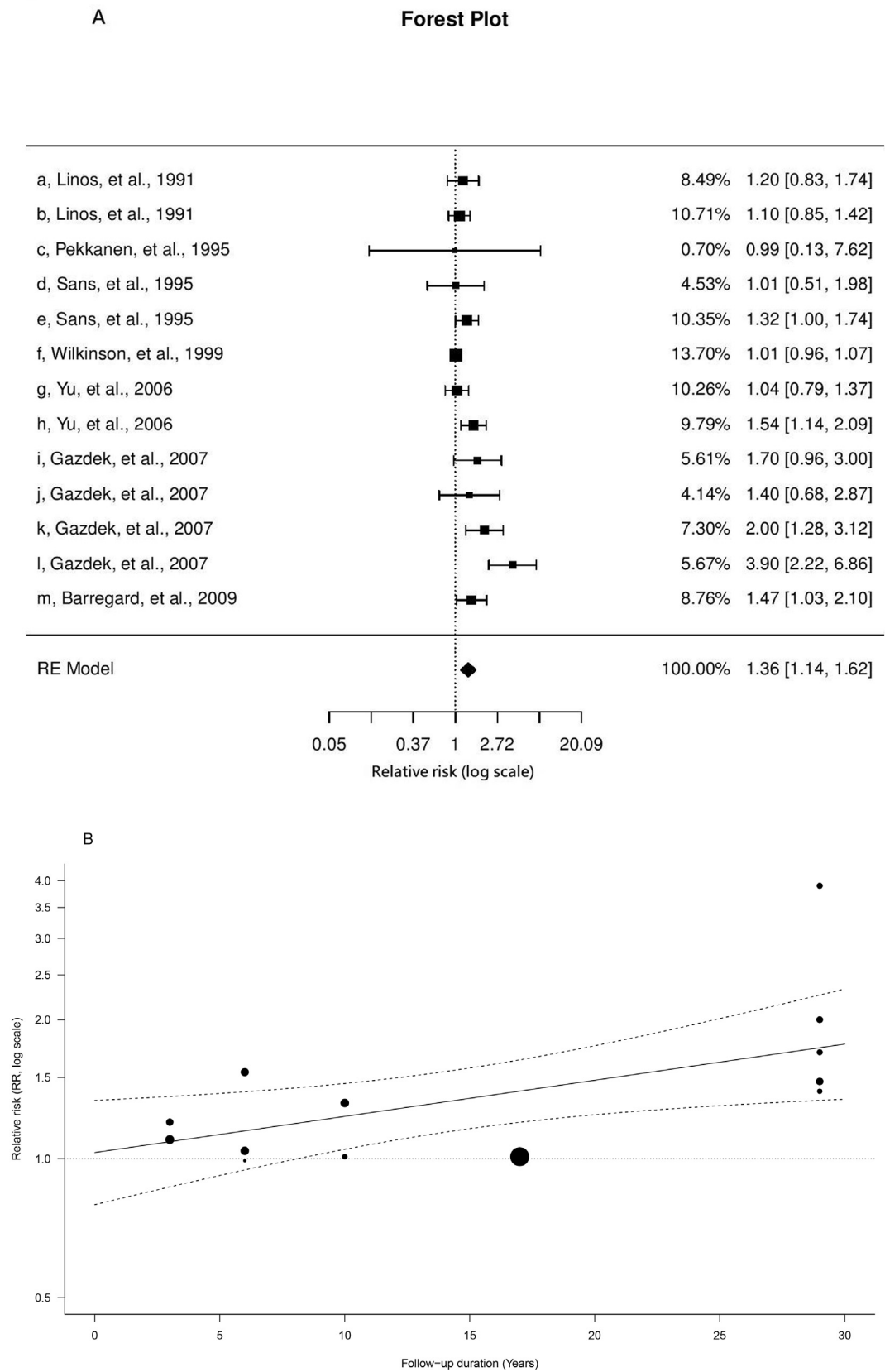


Fig. 2. Summarized relative risks of leukemia across different follow-up years. Panel A: Forest plot of included studies on the relative risks of leukemia. Each dot represents distinctive estimate of one study populations. Panel B: Meta-regression of estimated relative risks of leukemia in different follow-up years.

Table 1
Basic characteristics of 13 study populations among 7 included studies.

Authors	ID	Study population	Age distribution (years)	Number of leukemia cases	Threshold of distance-to-PICs (radius in km)	Effect estimates (95% CI)	Exposed groups (Potential chemicals)	Reference controls	Outcome: types of leukemia	Location	PICs operation years	Follow-up years	Adjusted covariates		
Linos et al.	A	Mainly white population	All	248	3.2	RR 1.20 (0.90–1.90)	In-person interview (Not provided)	Frequency-matched controls from same states	Leukemia, ALL, AML, CLL, CML	Iowa and Minnesota, USA	1955–2018	1980–1983	age, state		
	B	Mainly white population	All	142	RR 1.10 (0.90–1.50)										
Pekkanen et al.	C	Mainly white population	All	2	8	RR 0.99 (0.13–7.70)	Central population registry (SO ₂ , CO ₂ , VOC, vinyl chloride monomer, styrene)	Finish general residents	Leukemia	Finland	1965–2019	1983–1989	age, sex, distance from oil refinery		
Sans et al.	D	Mainly white children population	0–14	10	7.5	SIR 1.01 (0.48–1.85)	Distance from PICs (Alcohols, styrene, olefins, benzene, vinyl chloride monomer, polyvinyl chloride)	Residents from Great Britain	Leukemia, HL, NHL	Baglan Bay plant area, UK	1963–2004	1974–1984	age,sex,SES		
	E	Mainly white population	15–64	51	SIR 1.32 (1.00–1.73)										
Wilkinson et al.	F	Mainly white population	All	1446	7.5	SIR 1.01 (0.96–1.07)	Residential post codes (Benzene, 1,3-Butadiene)	General residents from Great Britain	Leukemia, ALL, AML, CLL, CML, HL, NHL	Seven refinery sites in UK	early 1970–2019	England, Wales: 1974–1989; Scotland: 1975–1991	age,sex,SES		
Yu et al.	G	Asian children	0–19	131	3	OR 1.04 (0.79–1.38)	Exposure opportunity score (Acrylates, acrylonitrile-butadiene-styrene, Thermoplastic-elastomer, Methyl tertiary butyl ether, Polyethene)	Randomly selected subjects in the same city	Leukemia, ALL	Kaohsiung, Taiwan	1968–2015	1997–2003	age,sex		
	H	Asian young adults	20–29	40	OR 1.54 (1.14–2.09)										
Gazdek et al.	I	Mainly white population, males	All	42	7.5	ASRR 1.70 (0.98–3.06)	Living in the exploitation regions (Lead, mercury, hydrocarbons, BTEX, and PABS; possible benzene; nitrogenous compounds)	Croatian general population	Leukemia, ALL, AML, CML, CLL, HL, NHL	Djurdjevac, Croatia	1981–2000	1971–2000	age,sex		
	J	Mainly white population, females	All	35	ARSS 1.40 (1.20–5.06)										
	K	Mainly white population, males	All	75	ASRR 2.00 (1.28–3.12)	Koprivnica, Croatia									
	L	Mainly white population, females	All	67	ASRR 3.90 (2.22–6.87)										
Barregard et al.	M	Mainly white population	All	33	7	SIR 1.47 (1.01–2.07)	Passive sampler for levels of toxic chemicals (Benzene, ethylene, 1,3-butadiene, propene, alkanes, alkenes, aromatic hydrocarbons)	Residents in the same area but away from the refinery.	Leukemia	Lysekil, Sweden	1975–2019	1975–2004	age,sex		

PICs, petrochemical industrial complexes; CI, confidence interval; RR, relative risk; SIR, standardized incidence ratio; ASRR, age-standardized incidence rate ratio; ALL, acute lymphoblastic leukemia; AML, acute myeloid leukemia; CLL, chronic lymphocytic leukemia; CML, chronic myeloid leukemia; HL, Hodgkin lymphoma; NHL, non-Hodgkin lymphoma; SES, socioeconomic status; ICD, International Classification of Diseases.

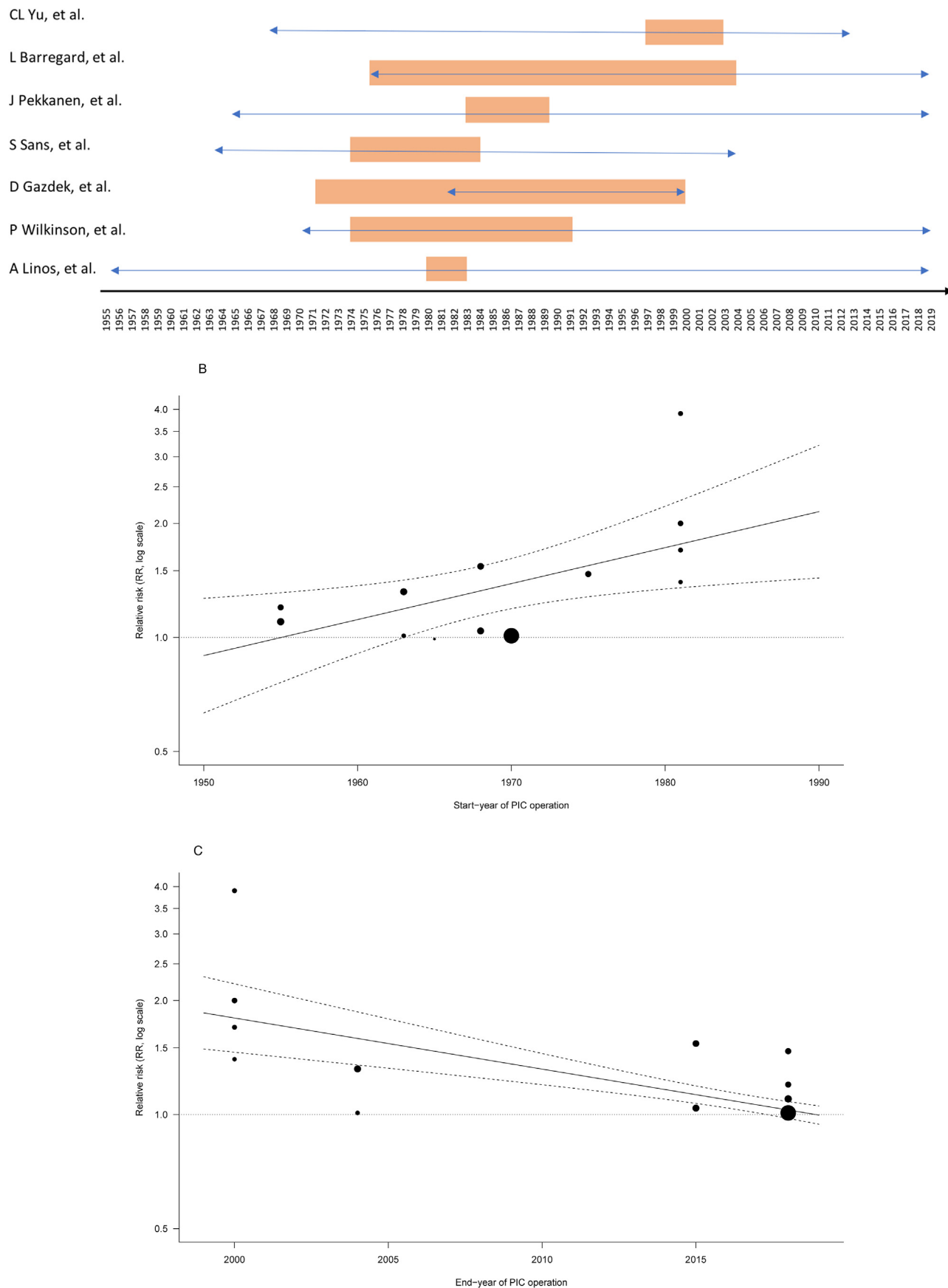


Fig. 3. Meta-regression of the relative risks of leukemia across different time periods of petrochemical industrial complex (PIC) operation and latency. Panel A: Illustration of the operation periods of PICs and follow-up periods of included studies. Panel B: Meta-regression of relative risks of leukemia across different start-year of PIC operation. Panel C: Meta-

Table 2

The pooled relative risks (RRs) of different subtypes of leukemia and leukemia stratified by different characteristics.

Characteristics	n	Pooled RR (95% CI)	I ² (%)	Tau	Q test	p-value
Overall	13	1.36 (1.14–1.62)	73.0	0.0230	0.257	0.001
Subtypes of leukemia						
ALL	7	1.50 (0.97–1.69)	64.4	0.3654	0.0217	0.069
AML	7	1.61 (1.12–2.31)	50.6	0.2986	0.0128	0.010
CLL	7	1.85 (1.01–6.42)	92.0	0.7274	<0.0001	0.048
CML	6	1.09 (0.96–1.25)	<0.1	0.0025	0.1097	0.189
HL	7	0.98 (0.90–1.07)	<0.1	0.0012	0.2070	0.684
NHL	7	1.01 (0.95–1.06)	<0.1	0.0009	0.0686	0.816
Reported effect measures						
RR	3	1.13 (0.92–1.39)	0	0	0.9240	0.924
OR	2	1.25 (0.96–1.64)	20.6	0.1279	0.0617	0.102
SIR/ASRR	8	1.52 (1.16–1.98)	78.8	0.3138	<0.0001	0.003
Sex						
Males	2	2.46 (1.22–4.97)	58.1	0.3885	0.0281	0.012
Females	2	1.88 (1.32–2.67)	0	0	0.6595	0.001
Both	9	1.16 (1.03–1.31)	40.2	0.1055	0.0639	0.015
Age						
Less than 30 years	3	1.22 (0.96–1.56)	27.1	0.1143	0.1475	0.108
All ages	10	1.43 (1.15–1.79)	77.8	0.2803	<0.0001	0.002
Location						
Asia	2	1.25 (0.96–1.64)	42.6	0.1279	0.0617	0.102
North America	2	1.13 (0.92–1.40)	0	0.3096	<0.0001	0.253
Europe	9	1.50 (1.15–1.96)	76.1	0	0.7063	0.003

n, numbers of study populations; RR, relative risk; CI, confidence interval; ALL, acute lymphoblastic leukemia; AML, acute myeloid leukemia; CLL, chronic lymphocytic leukemia; CML, chronic myeloid leukemia; HL, Hodgkin lymphoma; NHL, non-Hodgkin lymphoma; OR, odds ratio; SIR, standardized incidence ratio; ASRR, age-standardized rate ratio.

3.3. Quality and risk of bias assessment

The seven included studies were of high confidence on quality (scoring seven stars or more) by the NOQAS assessment (Supplementary Table 1). Discrepancies among the adjusted confounding covariates influenced the quality evaluation. Socioeconomic status, an important confounding *a priori*, was absent in two study populations (Gazdek et al., 2007; Yu et al., 2006b). Insufficient follow-up periods (i.e., less than 10 years) (Pekkanen et al., 1995b) and different follow-up periods for cases and controls (Gazdek et al., 2007) led to lower confidence on the NOQAS subsection (Supplementary Table 1). The detailed quantitative quality assessments were summarized in Appendices 1 (case-control studies) and 2 (cohort studies).

Sensitivity analysis by leave-one-out method demonstrated similarly summary RRs (pooled RRs ranging from 1.24 to 1.42 and I² ranging between 49.81 and 76.20), suggesting a valid effect estimation (Supplementary Table 2). The funnel plot (Supplementary Fig. 1) and Q-Q plot (Supplementary Fig. 2) showed no significant small-study effects because of the symmetric distribution in the funnel plot and a straight-line positioning in the Q-Q plot, respectively. The *p*-value of Begg's test is 0.51, suggesting no significant publication bias.

4. Discussion

This meta-analysis revealed that residential exposure to PICs was associated with an increased risk of incident leukemia by 36%. The effect was found to occur after 10 years of follow-up and slightly increased thereafter in meta-regression. The results were comparable with other studies on the association between benzene exposure and the risk of leukemia (Triebig, 2010).

Higher ambient levels of multiple chemicals (McMullin et al., 2018), which are commonly measured close to PICs, are associated with higher risks of leukemia, with different levels of evidence. For example, the causal association between benzene exposure and the development of leukemia has been well recognized (Snyder, 2012), as well as other volatile organic compounds (Dutta et al., 2018) and ionizing radiation (Terry et al., 2005). Living close to PICs is prone to expose to the substance through pathways of either air (Cetin et al., 2003), soil (Nadal et al., 2004), and drinking water (Wu et al., 2011). Our results indicate a higher risk of leukemia within a maximal distance of 8 km to PIC. As relevant to other distance-based analysis (Brender et al., 2011), higher risks of acute respiratory diseases, asthma, conjunctivitis and dermatitis were also observed among residents living within 10 km from industrial complexes (Al-Wahaibi and Zeka, 2015). Our results suggesting the association between distance to PICs and the risk of leukemia was in concordance with WHO report (Mudu and Martuzzi, 2014). An increased risk of leukemia was also noticed as the follow-up duration extended. The observation reflects a potential cumulative effect of exposure associated with increased risk of leukemia.

Exposure to PIC might be associated with different effects on subtypes of leukemia. Meta-regression showed increased risks of AML and CLL, which are common types of leukemia in the elderly. There are some animal and human observational studies revealing that PIC related toxic chemicals (e.g., benzene, toluene, ethylbenzene and xylene (BTEX), and other VOCs) could be associated with gene expression alteration and oxidative DNA damage (Andreoli et al., 1997) and subsequent leukemia. Moreover, growing studies supported the relationship between industrial chemical-induced epigenetic modifications and leukemia (Vivian Silva Kahl and Da Silva, 2019). For example, benzene exposure is associated

regression of relative risks of leukemia across different end-year of PIC operation. (Each dot represents one distinctive effect estimate of study population. The size of dots reflects the weight given in the dose-response regression. The black line and two dashed lines represent the relative risks, upper and lower bounds of confidence intervals, respectively. The horizontal line indicates relative risk of 1.)

with dysfunction of epigenetic modifiers histone deacetylase (HDAC) and DNA methyltransferase 3A (DNMT3A) (Abdel-Wahab and Levine, 2013), which could lead to early development of AML (Mizuno et al., 2001). Elder people are especially vulnerable to the harmful effect of the aforementioned PIC-related chemicals. This is consistent to the non-significant risk of leukemia among those aged below 30 in our analysis. One meta-analysis investigating exposure to benzene and the risk of leukemia subtypes revealed similar conclusions upon increased risks of AML and CLL (Khalade et al., 2010).

Our results found that the relative risks of leukemia varied across different calendar times of PICs operation, in accordance with the demand and regulations of benzene globally. The later starting time to operate PICs was associated with a higher risk of leukemia during 60's to 80's; while the later closing year of PICs with a lower risk after 2000. Worldwide, the demands of benzene and its products increased steeply from 1950 to 1975, with an average growth of 9% per year (Hodges, 2013). The US Occupational Safety and Health Administration acknowledged benzene and its metabolites as a carcinogen and issued the second regulation in 1987, after the first one was vacated from the supreme court in 1978 (Feitshans, 1989). Along with other economic factors, stricter regulations on benzenes and benzene metabolites were imposed across multiple countries worldwide (Feitshans, 1989; Park et al., 2015; Tuomi et al., 2018). For example, Finland has implemented environmental legislations for decades (such as Environmental Protection Acts since 1994) to regulate toxic wastes release into air, water and soils from PICs (Environment Mot, 2019). Similarly, several environmental laws and regulations were imposed (e.g., Toxic Chemical Substances Control Act in 1987, Air Pollution Control Act, first in 1975 and latest amended in 2006) and became more strictly in Taiwan over time (Industrial Development Bureau, 2007). With stringent regulations forcing existing PICs to release lower levels of toxic chemicals, the risk of leukemia associated with residential exposure to PICs has diminished in recent years, in concordance with our observations.

There were still a few potential limitations in this meta-analysis. First, the information about chemicals manufactured in those PICs was limited. There was no individual data regarding the exposed chemical concentrations and duration. Although the information at an individual level could help explain the mechanism of disease development, the study aimed to investigate the residential (occupational plus environmental) risks of leukemia among residents in the vicinity of PICs; thus, the causal relationship between exposure to certain PICs-related chemicals and disease incidence was not the focus of interest. Second, there was inconsistency in the definition of residential exposure to PICs. The distance to PICs used as proximity of exposure ranged from 0.8 km (Linos et al., 1991) to 8 km (Sans et al., 1995). There might have misclassification of exposed and unexposed groups, as the unexposed population in a study with shorter distances to PICs might be included in the exposed group in another study that had a broader exposure area. The mixing of residential distances used for the definition of "exposed" may bias effect estimates toward the null. Besides, although we found a higher risk of leukemia for residents living as far as 8 km, we had limited knowledge of the residential risk of those who lived even further away. In addition, people exposed to PICs might have more frequent health screening, which may possibly lead to early detection of diseases. Finally, most studies did not consider meteorological (Pekkanen et al., 1995a; Sans et al., 1995) and geographical differences (Yu et al., 2006a; Barregard et al., 2009). However, since residents lived both upwind and downwind of PICs, such considerations could reasonably be regarded as non-differential, and our estimates could still be robust.

5. Contributions

CKL, YTH and STC contributed to the study design, data analysis, reporting results, data interpretation, and drafting of the manuscript. KDB contributed to the study design, literature reviews, data acquisition, data interpretation, and rewriting of the manuscript. BP contributed to literature reviews, data acquisition, and data interpretation. YGW contributed to the review of the manuscript. All authors gave final approval of the version to be submitted and any revised version.

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Declaration of competing interest

The authors have declared that no competing interests exist.

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Appendix 1

Qualitative Summary of Newcastle-Ottawa Scale for case-control study

Case-control study (N = 2) (Yu et al., 2006a; Linos et al., 1991).

The quality of selected case-control studies was assessed using the Newcastle-Ottawa scale. Overall, both studies were considered as "good quality" as one received eight (Yu et al., 2006a) and the other received nine (Linos et al., 1991) out of nine stars.

In the session of *Selection*, cases in both selected studies had been identified with independent. The diagnosis of cancer had been confirmed by medical professionals through pathological examination. The cases represented the population well since most cases from census data revealed their living distances from petrochemical plants. The controls were selected from the same community. Through the cancer registry, we were certain that those controls were free from leukemia.

In the session of *Comparability*, all studies controlled for potential confounding factors, such as sex and gender. One study (Linos et al., 1991) provided a thorough explanation as to why they chose and did not control for potential factors. The other study (Yu et al., 2006a) lost one star because they did not provide explanations for choosing or not choosing some common factors, such as socioeconomic status, which is likely an important confounding factor (Poole et al., 2006).

In the session of *Exposure*, both studies had reliable data concerning the exposure record. The exposure had been measured using rigorous methods, such as geographic coding (Yu et al., 2006a; Linos et al., 1991), prevailing wind direction models (Yu et al., 2006a), and the exposure opportunity score (Yu et al., 2006a). The method of ascertainment of exposure was used in both cases and controls. Since the outcome was derived from national cancer registries directly, the universal coverage of citizens guaranteed equally high response rates for both the exposed and non-exposed.

In summary, the methods and study designs for both case-control studies were rigorous and solid. They had both reliable sources of cases and controls and adequate measures of exposure and outcomes to be considered of high quality.

Appendix 2

Qualitative Summary of Newcastle-Ottawa Scale for cohort study

Cohort study (N = 5) (Wilkinson et al., 1999; Pekkanen et al., 1995a; Barregard et al., 2009; Gazdek et al., 2007; Sans et al., 1995).

All of the five selected cohort studies received scores above seven, which designated them as being of “good quality” according to the scale. Three studies (Wilkinson et al., 1999; Barregard et al., 2009; Sans et al., 1995) received a full score of nine stars according to the assessment scale, one study (Pekkanen et al., 1995a) got eight stars and one (Gazdek et al., 2007) received eight stars out of nine.

In the section of *selection*, all studies earned full scores. The selected exposed groups represented the true populations living close to petrochemical industrial complexes (PIC) fairly. The non-exposed groups were selected from the same populations as exposed group but consisted of individuals resided at greater distances from the PICs. Exposure status was adequately assessed by identifying the individuals' residential distance from petrochemical plants. This data was derived from national census data, which served as a reliable source. The outcome of interest was carefully identified after the start of the study.

In the section of *Comparability*, most studies received two stars for comparability. They provided well-explained reasoning behind the selection of controlling factors. Only one study (Gazdek et al., 2007) lost a star in this section due to lack of an explanation for not controlling for socioeconomic status, a known potential confounding factor (Poole et al., 2006).

In the section of *Outcome*, all studies had the outcome of interest came from reliable sources using record linkage. Leukemia cases were all diagnosed by medical professionals according to standard measures. However, two studies had limited follow up years (less than 10 years) (Pekkanen et al., 1995a) or inconsistent follow-up periods for exposed vs. non-exposed group (Gazdek et al., 2007). By using national cancer registry data, all studies obtained highly satisfactory follow-up rates.

In summary, all five selected cohort studies received full or near-full scores in the quality assessment. They selected a representative population, had adequate measures of exposure, observed sufficient length of follow-up, and provided a satisfactory follow-up rates.

Appendix C. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envpol.2019.113476>.

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